

Cardiac Pathology Classification on the ACDC Dataset: A Comparison between a Tsetlin Machine and a Deep Neural Network

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1 Introduction

Cine cardiac magnetic resonance (CMR) is the reference technique for the assessment of ventricular volumes, function and myocardial mass. Many cardiomyopathies can be recognised from a small set of quantitative indices derived from the segmentation of the left ventricle (LV), right ventricle (RV) and myocardium (MYO) at the two extremes of the cardiac cycle, end-diastole (ED) and end-systole (ES) [1].

Deep neural networks are the standard choice for this kind of tabular classification, but their decisions are difficult to inspect, which is a limit in a clinical setting. The Tsetlin Machine [2] is an alternative learning algorithm that represents each class as a set of conjunctive clauses over Boolean literals (e.g. “*LV ejection fraction $\leq$ threshold AND LV end-diastolic volume $>$ threshold*”). Its decisions are therefore readable by a clinician.

The aim of this work is to compare the two approaches *under identical conditions*: same input features, same data split, same model-selection protocol and same evaluation metrics. We assess classification accuracy and also the calibration of the predicted probabilities, which is important when a model is used for clinical decision support.

2 Dataset

We use the Automated Cardiac Diagnosis Challenge (ACDC) dataset [1], acquired at the University Hospital of Dijon. It contains cine-CMR short-axis acquisitions of 150 patients, equally divided into five groups:

- **NOR** – normal subjects;
- **MINF** – previous myocardial infarction (LV ejection fraction $< 40\%$ and abnormal wall motion);
- **DCM** – dilated cardiomyopathy (LV end-diastolic volume $> 100 \text{ mL/m}^2$ and ejection fraction $< 40\%$);
- **HCM** – hypertrophic cardiomyopathy (LV ejection fraction $> 55\%$ and myocardial thickness $> 15 \text{ mm}$ in diastole);
- **RV** – abnormal right ventricle (RV volume $> 110 \text{ mL/m}^2$ or RV ejection fraction $< 40\%$).

For each patient the dataset provides the 4D cine sequence, the ED and ES frames with the expert manual segmentation (labels: 0 background, 1 RV cavity, 2 myocardium, 3 LV cavity), and an `Info.cfg` file with diagnosis, ED/ES frame indices, height and weight. We use the official split:

- **Training set:** patients 001–100 (20 per class), used for training and model selection;
- **Test set:** patients 101–150 (10 per class), used *only once* for the final evaluation.

In the test folder the classes are not ordered by patient ID. Labels and ED/ES frames are therefore always read from `Info.cfg` and never inferred from the ID.

3 Feature Extraction

Features are computed from the ground-truth segmentations at ED and ES, using the real voxel spacing stored in the NIfTI header. Volumes are expressed in mL, myocardial mass in g (density 1.05 g/mL) and thicknesses in mm. The body surface area is computed with the Mosteller formula [4], $BSA = \sqrt{h [\text{cm}] \cdot w [\text{kg}]/3600}$.

Myocardial wall thickness is estimated slice by slice. On each short-axis slice we compute the Euclidean distance transform of the myocardium mask and evaluate it on the skeleton of the mask. The local thickness is twice the distance from the skeleton to the closest boundary. For each phase we keep the maximum thickness, the mean thickness and the standard deviation of the slice-wise mean thickness. Table 1 lists the resulting 26 features.

Table 1: The 26 features extracted for each patient (shared by both models).

Group	Features	Unit
LV function	LV EDV, LV ESV, LV EF	mL, mL, –
RV function	RV EDV, RV ESV, RV EF	mL, mL, –
Myocardium	MYO mass at ED, MYO volume at ES	g, mL
Indexed (BSA)	LV EDV _i , LV ESV _i , RV EDV _i , RV ESV _i , MYO mass _i	mL/m ² , g/m ²
Ratios	RV/LV EDV, RV/LV ESV, MYO/LV at ED, MYO/LV at ES	–
Wall thickness	max, mean, std at ED; max, mean, std at ES	mm
Anthropometry	height, weight, BSA	cm, kg, m ²

Table 2 shows that the features have a clear physiological meaning. DCM has a very large indexed LV volume and a very low LV ejection fraction. HCM has the largest maximum wall thickness and a preserved ejection fraction. RV has the largest RV/LV volume ratio. MINF lies between DCM and NOR, which explains most of the errors discussed in Section 5.

Table 2: Mean $\pm$ standard deviation of selected features per class (training set, $n = 20$ per class).

Feature	DCM	HCM	MINF	NOR	RV
LV EDV _i (mL/m ²)	153.3 $\pm$ 31.6	67.2 $\pm$ 17.6	92.5 $\pm$ 25.7	67.8 $\pm$ 13.1	60.2 $\pm$ 18.5
LV EF	0.18 $\pm$ 0.07	0.67 $\pm$ 0.09	0.31 $\pm$ 0.08	0.60 $\pm$ 0.05	0.55 $\pm$ 0.06
RV EDV _i (mL/m ²)	94.2 $\pm$ 33.5	60.9 $\pm$ 14.1	63.4 $\pm$ 17.6	79.6 $\pm$ 17.5	111.9 $\pm$ 25.6
RV EF	0.32 $\pm$ 0.18	0.60 $\pm$ 0.11	0.56 $\pm$ 0.10	0.55 $\pm$ 0.08	0.31 $\pm$ 0.15
MYO mass _i (g/m ²)	92.2 $\pm$ 22.7	92.2 $\pm$ 28.4	65.8 $\pm$ 10.1	53.2 $\pm$ 12.2	43.4 $\pm$ 12.2
Max thickness ED (mm)	12.8 $\pm$ 1.2	21.7 $\pm$ 3.7	14.5 $\pm$ 2.5	12.4 $\pm$ 1.3	12.0 $\pm$ 1.6
RV/LV EDV	0.62 $\pm$ 0.19	0.93 $\pm$ 0.18	0.72 $\pm$ 0.22	1.17 $\pm$ 0.11	2.00 $\pm$ 0.69

4 Models and Pipelines

Figure 1 summarises the two pipelines. They share the feature extraction, the data split and the model-selection protocol. They differ only in the pre-processing of the features, which each model

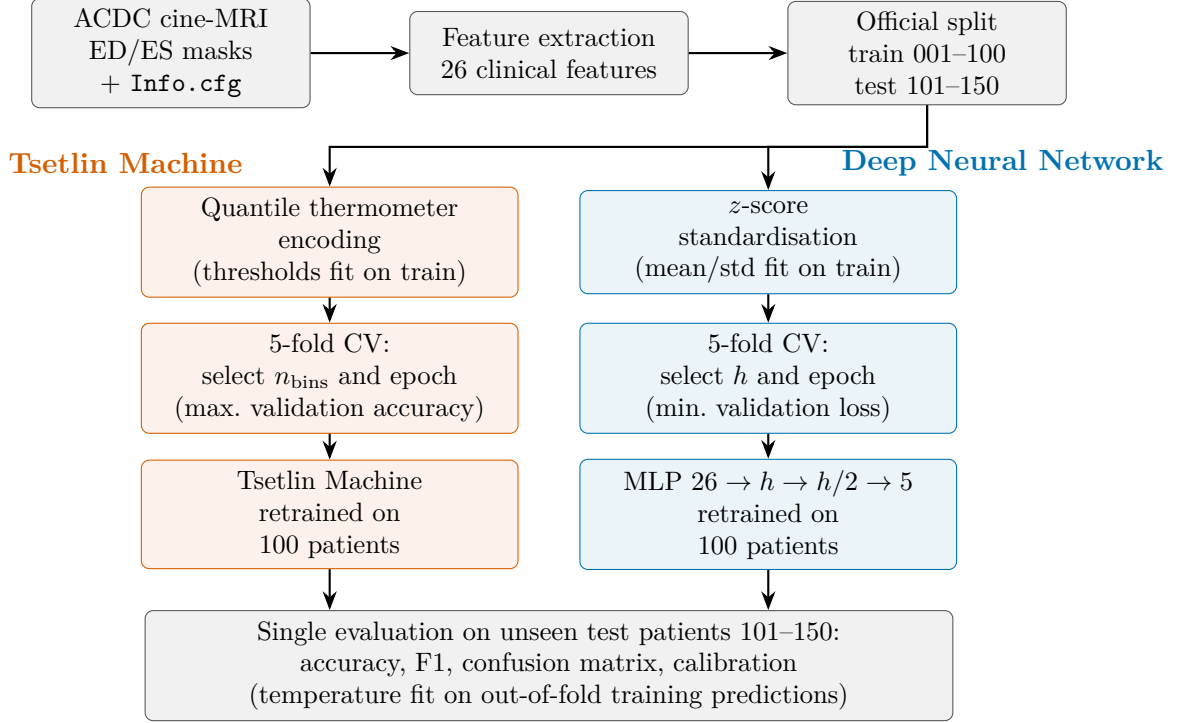


Figure 1: Pipelines of the two models. Grey blocks are shared; orange blocks are specific to the Tsetlin Machine and blue blocks to the DNN. Every quantity fitted on data (thresholds, mean/std, temperature) is estimated on training patients only.

needs in a different form, and in the classifier itself.

4.1 Tsetlin Machine

Booleanisation. The TM works on Boolean inputs, so every continuous feature x_j is converted with a *thermometer encoding*. Given thresholds $t_{j,1} < \dots < t_{j,B}$, the feature becomes the bits $[x_j > t_{j,1}], \dots, [x_j > t_{j,B}]$. The thresholds are the B inner quantiles of the feature distribution in the training data (inside cross-validation, in the training part of the fold). Each feature therefore gets its own thresholds, adapted to its range and distribution. With the selected $B = 5$ the input consists of 130 Boolean features and 260 literals (each feature and its negation).

Architecture. We use the multiclass TM with weighted clauses (`tmu` library). Each class has a bank of $C = 200$ clauses. Each clause is a conjunction of literals learned by a team of Tsetlin automata. Half of the clauses vote for the class and half against it. The class score is the (clipped) weighted sum of the clause outputs, and the prediction is the class with the highest score. The hyperparameters are $C = 200$ clauses per class, voting margin $T = 100$ and specificity $s = 5.0$.

Model selection. A 5-fold stratified cross-validation on patients 001–100 jointly selects the number of thresholds $B \in \{3, 5, 8, 10, 15\}$ and the number of training epochs (1–60). The criterion is the highest mean validation accuracy. The selected configuration is $B = 5$ and 48 epochs (Fig. 2). The final TM is then retrained on all 100 training patients.

4.2 Deep Neural Network

Pre-processing. Features are standardised to zero mean and unit variance with the statistics of the training data (again recomputed inside each cross-validation fold).

Architecture. A multilayer perceptron with two hidden layers:

$$\text{Linear}(26, h) \rightarrow \text{BatchNorm}(h) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.3) \\ \rightarrow \text{Linear}(h, h/2) \rightarrow \text{BatchNorm}(h/2) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.3) \rightarrow \text{Linear}(h/2, 5).$$

It is trained with the cross-entropy loss and the Adam optimiser (learning rate 10^{-3} , weight decay 10^{-4} , batch size 16).

Model selection. The capacity $h \in \{8, 16, 32, 64, 128, 256\}$ and the number of epochs (1–600) are chosen by 5-fold stratified cross-validation. The criterion is the *minimum mean validation loss*, an early-stopping criterion. Figure 3 shows the typical bias–variance trade-off. With $h = 8$ the network underfits (high training and validation loss). For $h \geq 64$ the training loss goes to zero while the validation loss increases (overfitting). The selected configuration is $h = 32$ (1,573 parameters) trained for 373 epochs (Fig. 4). The final network is retrained on all 100 training patients.

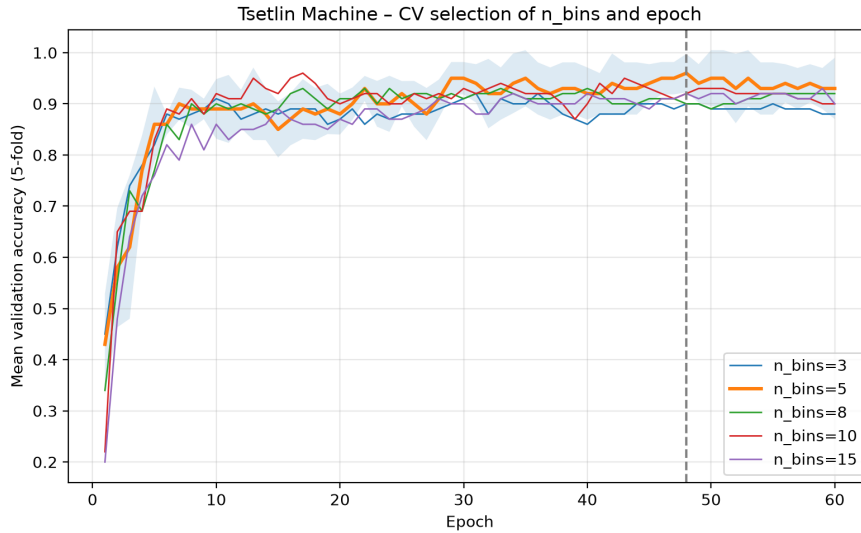


Figure 2: Tsetlin Machine: mean 5-fold validation accuracy per epoch for each number of thresholds B . The dashed line marks the selected epoch (48) for $B = 5$.

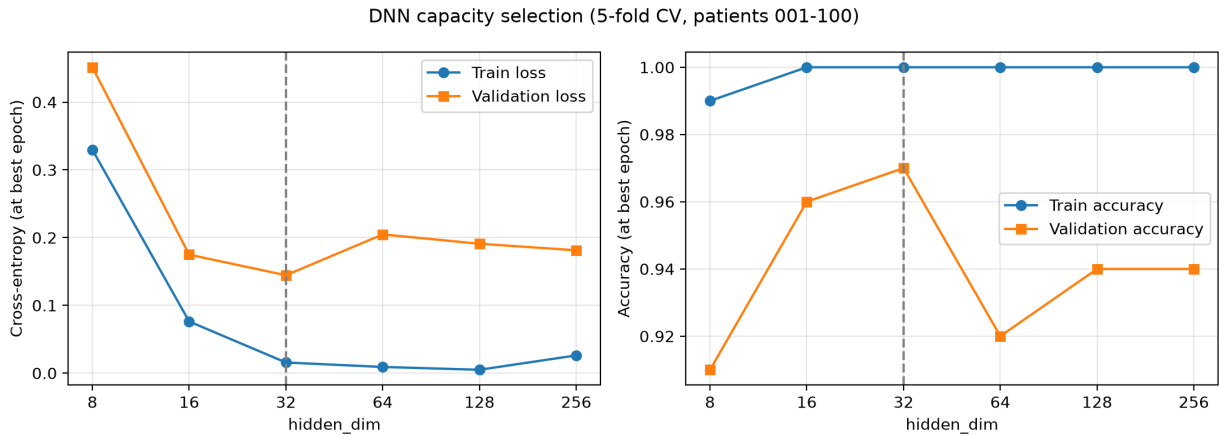


Figure 3: DNN capacity selection: training and validation cross-entropy and accuracy (at the best epoch of each configuration) as a function of the hidden dimension h .

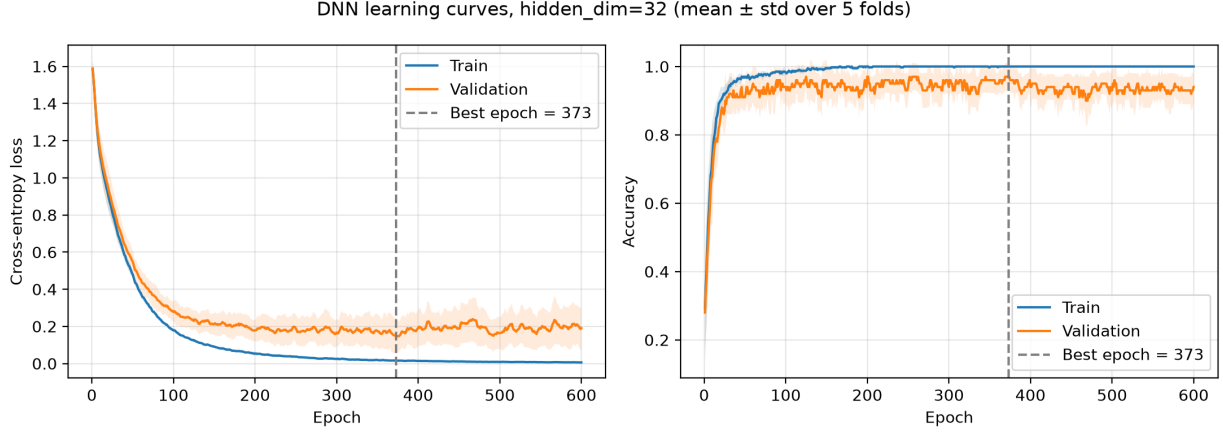


Figure 4: DNN learning curves for $h = 32$ (mean $\pm$ std over the 5 folds). The dashed line marks the epoch with minimum validation loss (373).

4.3 Probability calibration

For clinical use, the predicted probabilities should reflect the real chance of being correct. The DNN outputs logits, and the TM outputs class scores (clipped clause sums normalised by T). Both are turned into probabilities with a softmax with temperature τ , $p_k = \text{softmax}(z_k/\tau)$. The temperature is fitted by minimising the negative log-likelihood on the *out-of-fold* predictions of the training set (temperature scaling [3]), so the test set is never used. We report the expected calibration error (ECE, 5 equal-count bins) and the multiclass Brier score.

5 Results

5.1 Classification performance

Table 3 summarises the results and Fig. 5 shows the confusion matrices on the 50 test patients. Both models reach a similar cross-validation accuracy (0.96 for the TM, 0.97 for the DNN). On the test set, the DNN classifies 45/50 patients correctly (accuracy 0.90, macro-F1 0.90) and the TM 44/50 (accuracy 0.88, macro-F1 0.88).

Table 3: Performance on the test set (patients 101–150). P = precision, R = recall.

Class	Tsetlin Machine			DNN		
	P	R	F1	P	R	F1
DCM	0.82	0.90	0.86	0.75	0.90	0.82
HCM	0.83	1.00	0.91	1.00	0.90	0.95
MINF	0.89	0.80	0.84	0.88	0.70	0.78
NOR	1.00	0.70	0.82	0.91	1.00	0.95
RV	0.91	1.00	0.95	1.00	1.00	1.00
Accuracy	0.88			0.90		
Macro-F1	0.88			0.90		
CV accuracy (train)	0.96			0.97		
Selected config.	$B = 5$, 48 epochs			$h = 32$, 373 epochs		

Error analysis. The two models make different kinds of errors.

- **MINF** $\leftrightarrow$ **DCM** is the main source of errors for both models (3 errors for the TM, 4 for the DNN). Both classes have a dilated LV with reduced ejection fraction (Table 2). The clinical

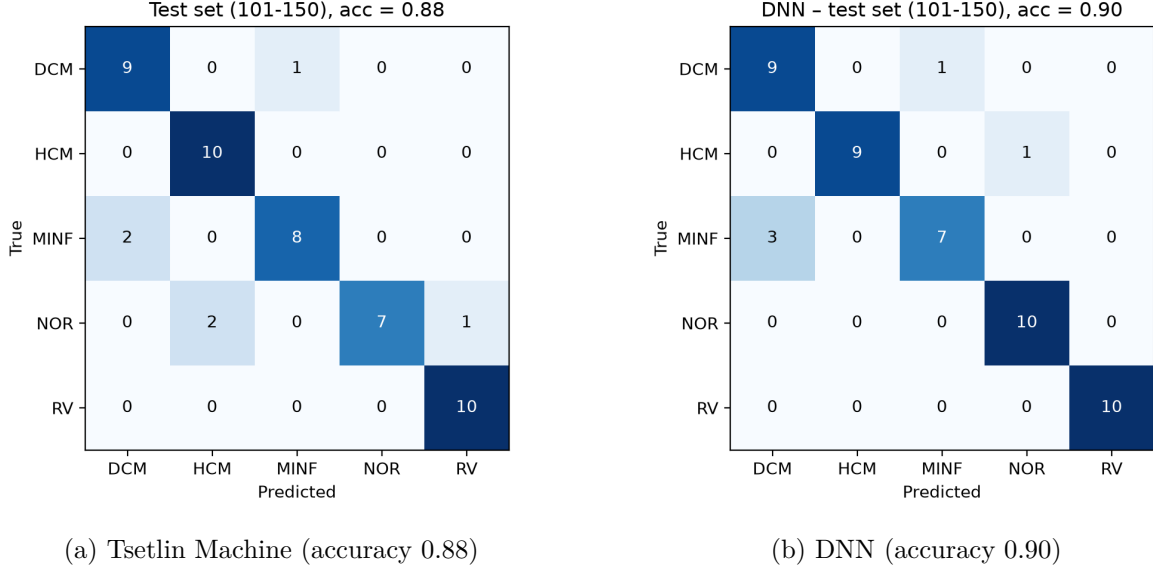


Figure 5: Confusion matrices on the test set (rows: true class, columns: predicted class).

criterion that separates them, regional wall-motion abnormality, is not captured by global volumetric features. Patients 135, 145 (MINF predicted as DCM) and 149 (DCM predicted as MINF) are misclassified by *both* models.

- The **TM** misclassifies 3 normal subjects (2 as HCM, 1 as RV). These are borderline values of wall thickness or RV/LV ratio that fall just beyond one of the learned thresholds. The thermometer encoding with 5 thresholds per feature produces sharp decision boundaries.
- The **DNN** recognises all NOR and RV patients, but one HCM patient is predicted as NOR.

Statistical comparison. On the test set, 2 patients are classified correctly only by the TM and 3 only by the DNN. The exact McNemar test gives $p = 1.0$, so the two models are not statistically different on this test set. We also repeated the whole pipeline (model selection included) with four random seeds. The test accuracy ranges from 0.82 to 0.90 for the TM (mean 0.865 ± 0.030) and from 0.84 to 0.90 for the DNN (mean 0.875 ± 0.022). The seed-to-seed variability is larger than the difference between the models.

Table 4: Test accuracy for different random seeds (full pipeline repeated, including model selection).

Model	seed 42	seed 1	seed 7	seed 123	mean $\pm$ std
Tsetlin Machine	0.88	0.82	0.86	0.90	0.865 ± 0.030
DNN	0.90	0.88	0.84	0.88	0.875 ± 0.022

5.2 Calibration

Figure 6 shows the reliability diagrams and Table 5 the calibration metrics. The two models behave very differently before calibration.

- The **TM** class scores, converted with a plain softmax ($\tau = 1$), give strongly *under-confident* probabilities (mean confidence around 0.5 for an accuracy of 0.88; ECE 0.37). The clause sums are bounded in $[-T, T]$, so their normalised range is narrow. Temperature scaling with the fitted $\tau = 0.205$ sharpens the distribution and reduces the ECE to 0.08 and the Brier score from 0.38 to 0.22.

- The **DNN** is already well calibrated (ECE 0.057, Brier 0.184). The fitted temperature ($\tau = 0.86$) is close to 1 and changes little. Thanks to early stopping on the validation loss, the network does not become over-confident.

After temperature scaling, the DNN still has a lower Brier score (0.18 vs 0.22). Its probabilities are therefore somewhat more informative, even though the accuracies are similar.

Table 5: Calibration metrics on the test set (lower is better). τ is fitted on out-of-fold training predictions.

	Tsetlin Machine		DNN	
	raw	scaled	raw	scaled
Temperature τ	1	0.205	1	0.86
ECE	0.367	0.082	0.057	0.049
Brier	0.379	0.222	0.184	0.185

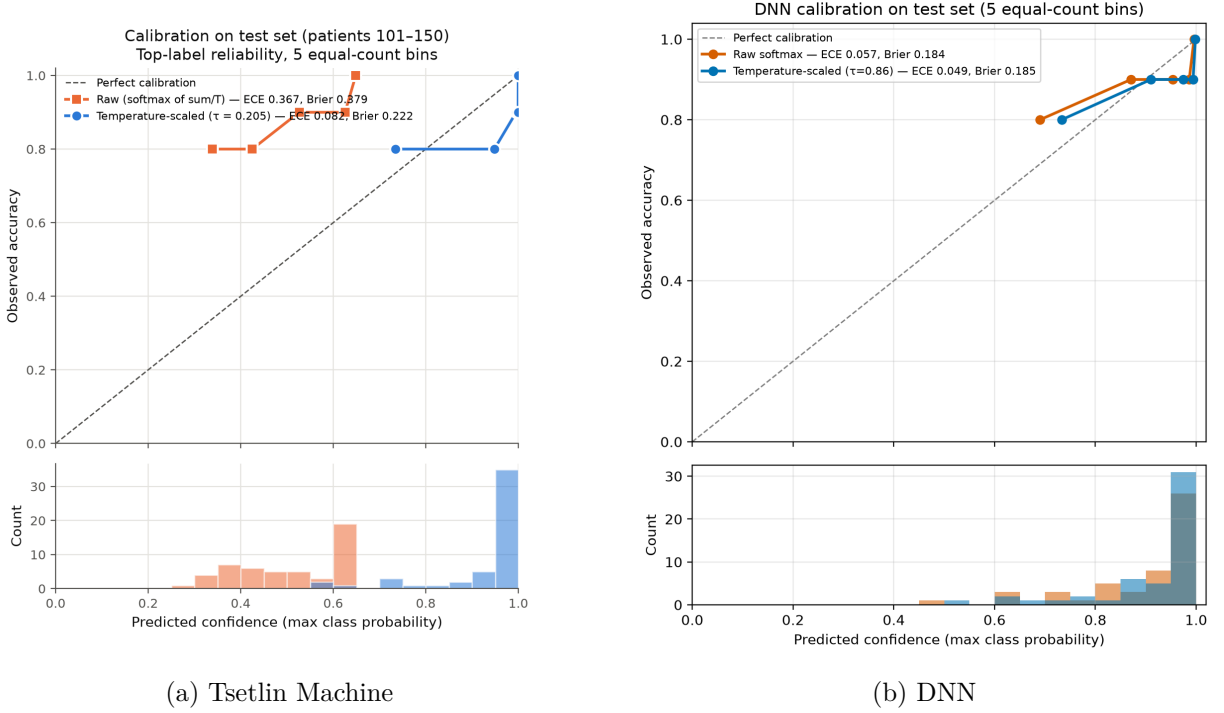


Figure 6: Top-label reliability diagrams on the test set (raw vs temperature-scaled) with the histogram of predicted confidences. With only 50 test patients the curves are noisy and should be read together with ECE and Brier score.

6 Conclusion

We compared a Tsetlin Machine and a deep neural network for the classification of five cardiac conditions on the ACDC dataset, using the same 26 clinical features extracted from the ED/ES segmentations and an identical, leakage-free protocol. The main findings are:

1. **Comparable accuracy.** The DNN reaches 0.90 and the TM 0.88 on the unseen test patients. The difference (one patient) is not statistically significant and is smaller than the variability across random seeds. Hand-crafted clinical features already separate the classes well, so a logic-based model can match a neural network on this task.

2. **Same main difficulty.** Both models mostly confuse previous infarction with dilated cardiomyopathy, because global volumetric indices cannot capture regional wall-motion abnormalities. Regional features, such as per-segment thickness or thickening following the AHA 17-segment model, are the most promising way to improve both models.
3. **Different calibration.** The DNN produces well-calibrated probabilities directly. The TM scores must be rescaled (temperature scaling) to be read as probabilities. After scaling, both are reasonably calibrated, with a slight advantage for the DNN.
4. **Interpretability.** The TM classifies with explicit logical rules over clinically meaningful thresholds (e.g. on ejection fraction, indexed volumes and wall thickness). This makes its decisions directly auditable, which the DNN does not offer. For a small, structured clinical dataset like ACDC, the TM is therefore a competitive and more transparent alternative.

References

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